

Supplemental: Formalized Lean 4 Proofs Local Softmax and Global Weights in Non-Boolean Event Structures

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Overview

This document contains the verbatim source code for the formal verification of the paper. The accompanying executable repository is archived on Zenodo at <https://doi.org>.

Contents

1	Project Configuration files	2
2	Core representation theorem	3
3	Gluing Logic	7
4	Pentagon Bounds Verification	10

1 Project Configuration files

lean-toolchain

```
leanprover/lean4:v4.30.0
```

lakefile.lean

```
1 import Lake
2 open Lake DSL
3
4 package SoftmaxVerification where
5
6 require mathlib from git
7   "https://github.com/leanprover-community/mathlib4.git" @ "v4.30.0"
8
9 @[default_target]
10 lean_lib SoftmaxVerification where
```

SoftmaxVerification.lean

```
1 import SoftmaxVerification.Core
2 import SoftmaxVerification.Gluing
3 import SoftmaxVerification.Pentagon
```

2 Core representation theorem

Core.lean

```
1 import Mathlib
2
3 open scoped BigOperators
4
5 namespace SoftmaxVerification
6
7 /--
8 A finite event structure in the sense used in the softmax paper.
9
10 `atoms` is the ambient finite set of atoms. Each context is itself a finite
11 set of atoms, and `context_subset` records that contexts do not contain
12 anything outside the ambient atom set.
13 -/
14 structure EventStructure (Atom : Type*) [DecidableEq Atom] where
15   atoms : Finset Atom
16   contexts : Finset (Finset Atom)
17   context_subset : forall C, C ∈ contexts -> C ⊆ atoms
18
19 /--
20 An admissible weight on an event structure: nonnegative on atoms and
21 normalized to one on every context.
22 -/
23 structure IsWeight {Atom : Type*} [DecidableEq Atom]
24   (L : EventStructure Atom) (p : Atom -> Real) : Prop where
25   nonneg : forall a, a ∈ L.atoms -> 0 ≤ p a
26   normalized : forall C, C ∈ L.contexts -> Finset.sum C p = 1
27
28 /-- Strict positivity on all atoms of the event structure. -/
29 def StrictlyPositive {Atom : Type*} [DecidableEq Atom]
30   (L : EventStructure Atom) (p : Atom -> Real) : Prop :=
31   forall a, a ∈ L.atoms -> 0 < p a
32
33 theorem strictlyPositive_nonneg {Atom : Type*} [DecidableEq Atom]
34   {L : EventStructure Atom} {p : Atom -> Real}
35   (hpos : StrictlyPositive L p) :
36   forall a, a ∈ L.atoms -> 0 ≤ p a := by
37   intro a ha
38   exact le_of_lt (hpos a ha)
39
40 /-- The normalizing denominator for positive coordinates on a context. -/
41 def denom {Atom : Type*} (q : Atom -> Real) (C : Finset Atom) : Real :=
42   Finset.sum C q
43
44 /-- Context-wise normalization of coordinates. -/
45 noncomputable def normalizedCoordinate {Atom : Type*}
```

```

46   (q : Atom -> Real) (C : Finset Atom) (a : Atom) : Real :=
47   q a / denom q C
48
49 /--
50 `q` represents the admissible weight `p` by normalized coordinates if the
51 coordinates are positive on atoms and context-wise normalization recovers `p`.
52 -/
53 def Represents {Atom : Type*} [DecidableEq Atom]
54   (L : EventStructure Atom) (p q : Atom -> Real) : Prop :=
55   (forall a, a ∈ L.atoms -> 0 < q a) ∧
56   forall C, C ∈ L.contexts -> forall a, a ∈ C ->
57     normalizedCoordinate q C a = p a
58
59 /--
60 Coordinate form of the paper's representation theorem.
61
62 Every strictly positive admissible weight represents itself: take the
63 unnormalized coordinate of an atom to be its weight.
64 -/
65 theorem positive_weight_represents_itself {Atom : Type*} [DecidableEq Atom]
66   {L : EventStructure Atom} {p : Atom -> Real}
67   (hp : IsWeight L p) (hpos : StrictlyPositive L p) :
68   Represents L p p := by
69   constructor
70   · exact hpos
71   · intro C hC a ha
72     unfold normalizedCoordinate denom
73     rw [hp.normalized C hC]
74     simp
75
76 /--
77 The same representation after multiplying all coordinates by a positive
78 constant. This is the basic gauge freedom of normalized coordinates.
79 -/
80 theorem positive_weight_scaled_coordinates {Atom : Type*} [DecidableEq Atom]
81   {L : EventStructure Atom} {p : Atom -> Real}
82   (hp : IsWeight L p) (hpos : StrictlyPositive L p)
83   {alpha : Real} (halpha : 0 < alpha) :
84   Represents L p (fun a => alpha * p a) := by
85   constructor
86   · intro a ha
87     exact mul_pos halpha (hpos a ha)
88   · intro C hC a ha
89     unfold normalizedCoordinate denom
90     have hsum : Finset.sum C (fun b => alpha * p b) = alpha := by
91       calc
92         Finset.sum C (fun b => alpha * p b) = alpha * Finset.sum C p := by
93           rw [Finset.mul_sum]
94         _ = alpha * 1 := by
95           rw [hp.normalized C hC]

```

```

96     _ = alpha := by
97       ring
98       rw [hsum]
99       field_simp [ne_of_gt halpha]
100
101 /--
102 Link-function version of the representation theorem.
103
104 Instead of formalizing an inverse function immediately, this theorem states the
105 exact range condition used in the paper: if every scaled value `alpha * p a`
106 has a preimage under the link `g`, then choosing those preimages as scores
107 gives the desired generalized-softmax representation.
108 -/
109 theorem link_representation_of_scaled_range
110   {Atom Score : Type*} [DecidableEq Atom] [Nonempty Score]
111   {L : EventStructure Atom} {p : Atom -> Real}
112   (g : Score -> Real)
113   (hp : IsWeight L p)
114   {alpha : Real} (halpha : 0 < alpha)
115   (hpre : forall a, a ∈ L.atoms -> exists u : Score, g u = alpha * p a) :
116   exists u : Atom -> Score,
117     forall C, C ∈ L.contexts -> forall a, a ∈ C ->
118       normalizedCoordinate (fun b => g (u b)) C a = p a := by
119   classical
120   let u : Atom -> Score := fun a =>
121     if ha : a ∈ L.atoms then
122       Classical.choose (hpre a ha)
123     else
124       Classical.choice (inferInstance : Nonempty Score)
125   refine {u, ?}_
126   intro C hC a ha
127   have hCsub : C ⊆ L.atoms := L.context_subset C hC
128   have hu : forall b, b ∈ C -> g (u b) = alpha * p b := by
129     intro b hb
130     have hb_atoms : b ∈ L.atoms := hCsub hb
131     dsimp [u]
132     rw [dif_pos hb_atoms]
133     exact Classical.choose_spec (hpre b hb_atoms)
134   unfold normalizedCoordinate denom
135   have hsum : Finset.sum C (fun b => g (u b)) = alpha := by
136     calc
137       Finset.sum C (fun b => g (u b)) = Finset.sum C (fun b => alpha * p b) :=
138         by
139           apply Finset.sum_congr rfl
140           intro b hb
141           exact hu b hb
142       _ = alpha * Finset.sum C p := by
143         rw [Finset.mul_sum]
144       _ = alpha * 1 := by
145         rw [hp.normalized C hC]

```

```
145   _ = alpha := by
146     ring
147     change g (u a) / Finset.sum C (fun b => g (u b)) = p a
148     rw [hu a ha, hsum]
149     field_simp [ne_of_gt halpha]
150
151 end SoftmaxVerification
```

3 Gluing Logic

Gluing.lean

```
1 import SoftmaxVerification.Core
2
3 open scoped BigOperators
4
5 namespace SoftmaxVerification
6
7 /-- A context-indexed family of local probabilities is nonnegative on
8 the atoms of each context. -/
9 def LocalNonnegative {Atom : Type*} [DecidableEq Atom]
10   (L : EventStructure Atom) (P : Finset Atom -> Atom -> Real) : Prop :=
11   forall C, C ∈ L.contexts -> forall a, a ∈ C -> 0 <= P C a
12
13 /-- A context-indexed family of local probabilities is normalized in
14 each context. -/
15 def LocalNormalized {Atom : Type*} [DecidableEq Atom]
16   (L : EventStructure Atom) (P : Finset Atom -> Atom -> Real) : Prop :=
17   forall C, C ∈ L.contexts -> Finset.sum C (P C) = 1
18
19 /-- Single-valuedness/no-disturbance for context-indexed local
20 probabilities. Shared atoms receive the same value in all contexts in
21 which they occur. -/
22 def SingleValuedOn {Atom : Type*} [DecidableEq Atom]
23   (L : EventStructure Atom) (P : Finset Atom -> Atom -> Real) : Prop :=
24   forall C, C ∈ L.contexts -> forall D, D ∈ L.contexts -> forall a,
25     a ∈ C -> a ∈ D -> P C a = P D a
26
27 /-- Local normalizing denominator for freely context-dependent
28 positive coordinates. -/
29 def localDenom {Atom : Type*}
30   (q : Finset Atom -> Atom -> Real) (C : Finset Atom) : Real :=
31   Finset.sum C (q C)
32
33 /-- Local softmax normalization for freely context-dependent positive
34 coordinates. -/
35 noncomputable def localCoordinate {Atom : Type*}
36   (q : Finset Atom -> Atom -> Real) (C : Finset Atom) (a : Atom) : Real :=
37   q C a / localDenom q C
38
39 /-- The paper's gluing condition, stated as single-valuedness of the
40 locally normalized coordinates. -/
41 theorem gluing_condition_for_free_scores {Atom : Type*} [DecidableEq Atom]
42   {L : EventStructure Atom} {q : Finset Atom -> Atom -> Real} :
43   SingleValuedOn L (localCoordinate q) <=>
44     forall C, C ∈ L.contexts -> forall D, D ∈ L.contexts -> forall a,
45       a ∈ C -> a ∈ D ->
```

```

46     localCoordinate q C a = localCoordinate q D a := by
47     rfl
48
49     /-- Algebraic form of the normalizer-ratio condition for one shared
50     atom. This is the scalar calculation behind Eq. (4.3) in the paper. -/
51     theorem normalized_eq_iff_ratio
52       {qC qD ZC ZD : Real}
53       (hqD : qD ≠ 0) (hZC : ZC ≠ 0) (hZD : ZD ≠ 0) :
54       qC / ZC = qD / ZD <=> ZC / ZD = qC / qD := by
55     constructor
56     · intro h
57       field_simp [hZC, hZD] at h
58       field_simp [hZD, hqD]
59       nlinarith
60     · intro h
61       field_simp [hZD, hqD] at h
62       field_simp [hZC, hZD]
63       nlinarith
64
65     /-- A noncomputable glued global weight: choose one context containing
66     the atom, if such a context exists, and otherwise use zero. The choice
67     is harmless under `SingleValuedOn`. -/
68     noncomputable def gluedWeight {Atom : Type*} [DecidableEq Atom]
69       (L : EventStructure Atom) (P : Finset Atom -> Atom -> Real) :
70       Atom -> Real :=
71     fun a =>
72       if h : exists C, C ∈ L.contexts ∧ a ∈ C then
73         P (Classical.choose h) a
74       else
75         0
76
77     theorem gluedWeight_eq_of_mem {Atom : Type*} [DecidableEq Atom]
78       {L : EventStructure Atom} {P : Finset Atom -> Atom -> Real}
79       (hsingle : SingleValuedOn L P)
80       {C : Finset Atom} (hC : C ∈ L.contexts) {a : Atom} (ha : a ∈ C) :
81       gluedWeight L P a = P C a := by
82     classical
83     unfold gluedWeight
84     have hex : exists D, D ∈ L.contexts ∧ a ∈ D := {C, hC, }ha
85     rw [dif_pos hex]
86     have hchoose := Classical.choose_spec hex
87     exact hsingle (Classical.choose hex) hchoose.1 C hC a hchoose.2 ha
88
89     /-- Proposition 5.1 in finite form: every single-valued family of
90     locally normalized distributions defines an admissible global weight. -/
91     theorem local_distributions_glue_to_weight {Atom : Type*} [DecidableEq Atom]
92       {L : EventStructure Atom} {P : Finset Atom -> Atom -> Real}
93       (hnonneg : LocalNonnegative L P)
94       (hnorm : LocalNormalized L P)
95       (hsingle : SingleValuedOn L P) :

```



```

96   exists p : Atom -> Real,
97     IsWeight L p ∧
98     forall C, C ∈ L.contexts -> forall a, a ∈ C -> p a = P C a := by
99   classical
100  refine (gluedWeight L P, ?_, ?)_
101  · constructor
102  · intro a ha
103    unfold gluedWeight
104    by_cases hex : exists C, C ∈ L.contexts ∧ a ∈ C
105    · rw [dif_pos hex]
106      have hchoose := Classical.choose_spec hex
107      exact hnonneg (Classical.choose hex) hchoose.1 a hchoose.2
108    · rw [dif_neg hex]
109  · intro C hC
110    calc
111      Finset.sum C (gluedWeight L P) = Finset.sum C (P C) := by
112        apply Finset.sum_congr rfl
113        intro a ha
114        exact gluedWeight_eq_of_mem hsingle hC ha
115      _ = 1 := hnorm C hC
116  · intro C hC a ha
117    exact gluedWeight_eq_of_mem hsingle hC ha
118
119 end SoftmaxVerification

```

4 Pentagon Bounds Verification

Pentagon.lean

```
1 import SoftmaxVerification.Gluing
2
3 open scoped BigOperators
4
5 namespace SoftmaxVerification
6
7 /-- The ten atoms of the pentagon logic. -/
8 inductive PentagonAtom where
9   | a0 | a1 | a2 | a3 | a4
10  | x0 | x1 | x2 | x3 | x4
11  deriving DecidableEq, Fintype
12
13 open PentagonAtom
14
15 def pentagonC0 : Finset PentagonAtom := {a0, a1, x0}
16 def pentagonC1 : Finset PentagonAtom := {a1, a2, x1}
17 def pentagonC2 : Finset PentagonAtom := {a2, a3, x2}
18 def pentagonC3 : Finset PentagonAtom := {a3, a4, x3}
19 def pentagonC4 : Finset PentagonAtom := {a4, a0, x4}
20
21 def pentagonContexts : Finset (Finset PentagonAtom) :=
22   {pentagonC0, pentagonC1, pentagonC2, pentagonC3, pentagonC4}
23
24 /-- The pentagon/pentagram event structure from the paper. -/
25 def pentagon : EventStructure PentagonAtom where
26   atoms := Finset.univ
27   contexts := pentagonContexts
28   context_subset := by
29     intro C hC a ha
30     simp
31
32 /-- The cyclic atoms `a_i`. -/
33 def pentagonCyclicAtoms : Finset PentagonAtom := {a0, a1, a2, a3, a4}
34
35 def pentagonCyclicSum (p : PentagonAtom → Real) : Real :=
36   Finset.sum pentagonCyclicAtoms p
37
38 /-- The pentagon half-weight: cyclic atoms have weight `1/2`, midpoint
39 atoms have weight `0`. -/
40 noncomputable def pentagonHalfWeight : PentagonAtom → Real
41   | a0 | a1 | a2 | a3 | a4 => 1 / 2
42   | x0 | x1 | x2 | x3 | x4 => 0
43
44 theorem pentagon_halfWeight_isWeight : IsWeight pentagon pentagonHalfWeight :=
45   by
```

```

45 constructor
46   · intro a ha
47     cases a <=> norm_num [pentagonHalfWeight]
48   · intro C hC
49     simp [pentagon, pentagonContexts] at hC
50     rcases hC with hC | hC | hC | hC | hC
51     · subst C
52       simp [pentagonC0, pentagonHalfWeight]
53       norm_num
54     · subst C
55       simp [pentagonC1, pentagonHalfWeight]
56       norm_num
57     · subst C
58       simp [pentagonC2, pentagonHalfWeight]
59       norm_num
60     · subst C
61       simp [pentagonC3, pentagonHalfWeight]
62       norm_num
63     · subst C
64       simp [pentagonC4, pentagonHalfWeight]
65       norm_num
66
67 theorem pentagon_halfWeight_cyclicSum :
68   pentagonCyclicSum pentagonHalfWeight = 5 / 2 := by
69   simp [pentagonCyclicSum, pentagonCyclicAtoms, pentagonHalfWeight]
70   norm_num
71
72 /-- A two-valued admissible weight. -/
73 def IsTwoValued {Atom : Type*} [DecidableEq Atom]
74   (L : EventStructure Atom) (v : Atom -> Real) : Prop :=
75   IsWeight L v ∧ forall a, a ∈ L.atoms -> v a = 0 ∨ v a = 1
76
77 private theorem pentagon_adjacent_sum_le_one
78   {v : PentagonAtom -> Real} (hv : IsTwoValued pentagon v) :
79   v a0 + v a1 <= 1 ∧
80   v a1 + v a2 <= 1 ∧
81   v a2 + v a3 <= 1 ∧
82   v a3 + v a4 <= 1 ∧
83   v a4 + v a0 <= 1 := by
84   have hx0 : 0 <= v x0 := hv.1.nonneg x0 (by simp [pentagon])
85   have hx1 : 0 <= v x1 := hv.1.nonneg x1 (by simp [pentagon])
86   have hx2 : 0 <= v x2 := hv.1.nonneg x2 (by simp [pentagon])
87   have hx3 : 0 <= v x3 := hv.1.nonneg x3 (by simp [pentagon])
88   have hx4 : 0 <= v x4 := hv.1.nonneg x4 (by simp [pentagon])
89   have h0 := hv.1.normalized pentagonC0 (by simp [pentagon, pentagonContexts])
90   have h1 := hv.1.normalized pentagonC1 (by simp [pentagon, pentagonContexts])
91   have h2 := hv.1.normalized pentagonC2 (by simp [pentagon, pentagonContexts])
92   have h3 := hv.1.normalized pentagonC3 (by simp [pentagon, pentagonContexts])
93   have h4 := hv.1.normalized pentagonC4 (by simp [pentagon, pentagonContexts])
94   simp [pentagonC0] at h0

```

```

95 simp [pentagonC1] at h1
96 simp [pentagonC2] at h2
97 simp [pentagonC3] at h3
98 simp [pentagonC4] at h4
99 constructor
100 · nlinarith
101 constructor
102 · nlinarith
103 constructor
104 · nlinarith
105 constructor
106 · nlinarith
107 · nlinarith
108
109 /-- Every two-valued pentagon weight assigns value one to at most two
110 cyclic atoms. -/
111 theorem pentagon_twoValued_cyclicSum_le_two
112   {v : PentagonAtom -> Real} (hv : IsTwoValued pentagon v) :
113   pentagonCyclicSum v <= 2 := by
114   have hadj := pentagon_adjacent_sum_le_one hv
115   rcases hadj with (h01, h12, h23, h34, )h40
116   have ha0 := hv.2 a0 (by simp [pentagon])
117   have ha1 := hv.2 a1 (by simp [pentagon])
118   have ha2 := hv.2 a2 (by simp [pentagon])
119   have ha3 := hv.2 a3 (by simp [pentagon])
120   have ha4 := hv.2 a4 (by simp [pentagon])
121   rcases ha0 with ha0 | ha0 <=>
122   rcases ha1 with ha1 | ha1 <=>
123   rcases ha2 with ha2 | ha2 <=>
124   rcases ha3 with ha3 | ha3 <=>
125   rcases ha4 with ha4 | ha4 <=>
126   simp [pentagonCyclicSum, pentagonCyclicAtoms, ha0, ha1, ha2, ha3, ha4] <=>
127   nlinarith
128
129 /-- A finite classical mixture of two-valued pentagon weights obeys the
130 same cyclic inequality. -/
131 theorem pentagon_classical_mixture_cyclicSum_le_two
132   {ι : Type*} [Fintype ι]
133   {lambda : ι -> Real} {v : ι -> PentagonAtom -> Real}
134   {p : PentagonAtom -> Real}
135   (hlambda_nonneg : forall i, 0 <= lambda i)
136   (hlambda_sum : Finset.sum Finset.univ lambda = 1)
137   (hv : forall i, IsTwoValued pentagon (v i))
138   (hp : forall a, p a = Finset.sum Finset.univ (fun i => lambda i * v i a)) :
139   pentagonCyclicSum p <= 2 := by
140   have hsum :
141     pentagonCyclicSum p =
142       Finset.sum Finset.univ (fun i => lambda i * pentagonCyclicSum (v i)) :=
143       by
144         unfold pentagonCyclicSum

```

```

144   calc
145     Finset.sum pentagonCyclicAtoms p =
146       Finset.sum pentagonCyclicAtoms
147       (fun a => Finset.sum Finset.univ (fun i => lambda i * v i a)) := by
148       apply Finset.sum_congr rfl
149       intro a ha
150       exact hp a
151   _ = Finset.sum Finset.univ
152     (fun i => Finset.sum pentagonCyclicAtoms (fun a => lambda i * v i a
153       )) := by
154     rw [Finset.sum_comm]
155     _ = Finset.sum Finset.univ (fun i => lambda i * Finset.sum
156       pentagonCyclicAtoms (v i)) := by
157     apply Finset.sum_congr rfl
158     intro i hi
159     rw [Finset.mul_sum]
160   rw [hsum]
161   calc
162     Finset.sum Finset.univ (fun i => lambda i * pentagonCyclicSum (v i))
163     <= Finset.sum Finset.univ (fun i => lambda i * 2) := by
164     apply Finset.sum_le_sum
165     intro i hi
166     exact mul_le_mul_of_nonneg_left (pentagon_twoValued_cyclicSum_le_two (h
167       v i))
168     (hlambda_nonneg i)
169   _ = 2 := by
170   rw <[ Finset.sum_mul]
171   rw [hlambda_sum]
172   ring
173
174   theorem pentagon_halfWeight_exceeds_classical_bound :
175     2 < pentagonCyclicSum pentagonHalfWeight := by
176     rw [pentagon_halfWeight_cyclicSum]
177     norm_num
178
179   theorem pentagon_halfWeight_exceeds_kcbs_value :
180     Real.sqrt 5 < pentagonCyclicSum pentagonHalfWeight := by
181     rw [pentagon_halfWeight_cyclicSum]
182     rw [Real.sqrt_lt' (by norm_num : 0 < (5 : Real) / 2)]
183     norm_num
184
185   /- The positive generalized-softmax path used in the paper, written in
186   terms of the coordinate ratio `r`. -/
187   noncomputable def pentagonPathWeight (r : Real) : PentagonAtom -> Real
188   | a0 | a1 | a2 | a3 | a4 => 1 / (2 + r)
189   | x0 | x1 | x2 | x3 | x4 => r / (2 + r)
190
191   theorem pentagon_pathWeight_isWeight {r : Real} (hr : 0 <= r) :
192     IsWeight pentagon (pentagonPathWeight r) := by
193     have hden_pos : 0 < 2 + r := by linarith

```

```

191 constructor
192   · intro a ha
193     cases a <;> dsimp [pentagonPathWeight]
194     · exact div_nonneg zero_le_one hden_pos.le
195     · exact div_nonneg zero_le_one hden_pos.le
196     · exact div_nonneg zero_le_one hden_pos.le
197     · exact div_nonneg zero_le_one hden_pos.le
198     · exact div_nonneg zero_le_one hden_pos.le
199     · exact div_nonneg hr hden_pos.le
200     · exact div_nonneg hr hden_pos.le
201     · exact div_nonneg hr hden_pos.le
202     · exact div_nonneg hr hden_pos.le
203     · exact div_nonneg hr hden_pos.le
204   · intro C hC
205     simp [pentagon, pentagonContexts] at hC
206     rcases hC with hC | hC | hC | hC | hC
207     · subst C
208       simp [pentagonC0, pentagonPathWeight]
209       field_simp [ne_of_gt hden_pos]
210       ring_nf
211     · subst C
212       simp [pentagonC1, pentagonPathWeight]
213       field_simp [ne_of_gt hden_pos]
214       ring_nf
215     · subst C
216       simp [pentagonC2, pentagonPathWeight]
217       field_simp [ne_of_gt hden_pos]
218       ring_nf
219     · subst C
220       simp [pentagonC3, pentagonPathWeight]
221       field_simp [ne_of_gt hden_pos]
222       ring_nf
223     · subst C
224       simp [pentagonC4, pentagonPathWeight]
225       field_simp [ne_of_gt hden_pos]
226       ring_nf
227
228 theorem pentagon_pathWeight_cyclicSum (r : Real) :
229   pentagonCyclicSum (pentagonPathWeight r) = 5 / (2 + r) := by
230   simp [pentagonCyclicSum, pentagonCyclicAtoms, pentagonPathWeight]
231   ring_nf
232
233 theorem pentagon_path_exceeds_classical_iff {r : Real} (hr : 0 <= r) :
234   2 < pentagonCyclicSum (pentagonPathWeight r) <-> r < 1 / 2 := by
235   have hden_pos : 0 < 2 + r := by linarith
236   rw [pentagon_pathWeight_cyclicSum]
237   constructor
238   · intro h
239     field_simp [ne_of_gt hden_pos] at h
240     linarith

```

```

241   · intro h
242     field_simp [ne_of_gt hden_pos]
243     linarith
244
245 theorem pentagon_path_exceeds_kcbs_of_small {r : Real}
246   (hr : 0 ≤ r) (hsmall : r < Real.sqrt 5 - 2) :
247     Real.sqrt 5 < pentagonCyclicSum (pentagonPathWeight r) := by
248   have hden_pos : 0 < 2 + r := by linarith
249   have hsqrt_pos : 0 < Real.sqrt 5 := Real.sqrt_pos.2 (by norm_num)
250   have hlt : 2 + r < Real.sqrt 5 := by linarith
251   have hmul : Real.sqrt 5 * (2 + r) < Real.sqrt 5 * Real.sqrt 5 :=
252     mul_lt_mul_of_pos_left hlt hsqrt_pos
253   have hsq : Real.sqrt 5 * Real.sqrt 5 = 5 := by
254     rw ← [pow_two, Real.sq_sqrt]
255     norm_num
256   rw [pentagon_pathWeight_cyclicSum]
257   field_simp [ne_of_gt hden_pos]
258   nlinarith
259
260 theorem pentagon_pathWeight_strictlyPositive {r : Real} (hr : 0 < r) :
261   StrictlyPositive pentagon (pentagonPathWeight r) := by
262   intro a ha
263   have hden_pos : 0 < 2 + r := by linarith
264   cases a <|> dsimp [pentagonPathWeight]
265   · exact div_pos zero_lt_one hden_pos
266   · exact div_pos zero_lt_one hden_pos
267   · exact div_pos zero_lt_one hden_pos
268   · exact div_pos zero_lt_one hden_pos
269   · exact div_pos zero_lt_one hden_pos
270   · exact div_pos hr hden_pos
271   · exact div_pos hr hden_pos
272   · exact div_pos hr hden_pos
273   · exact div_pos hr hden_pos
274   · exact div_pos hr hden_pos
275
276 theorem pentagon_halfWeight_not_strictlyPositive :
277   ¬ StrictlyPositive pentagon pentagonHalfWeight := by
278   intro hpos
279   have hx := hpos x0 (by simp [pentagon])
280   norm_num [pentagonHalfWeight] at hx
281
282 theorem pentagon_pathWeight_zero_eq_halfWeight :
283   pentagonPathWeight 0 = pentagonHalfWeight := by
284   funext a
285   cases a <|> norm_num [pentagonPathWeight, pentagonHalfWeight]
286
287 end SoftmaxVerification

```